Lab 7: Fuzzing for Input Validation Bugs

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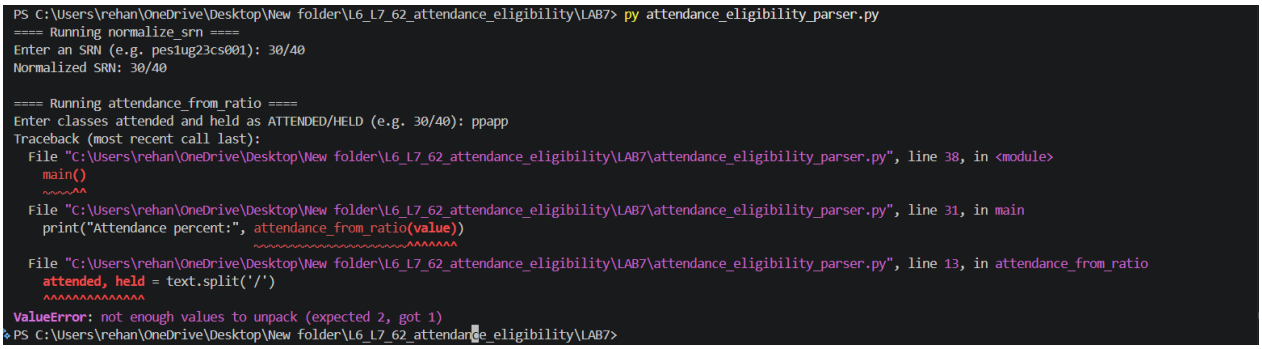
SECTION: 5J

Github repo:

https://github.com/rehanbawakhan/L6_L7_62_attendance_eligibility/tree/main/LAB7

Screenshot 1: Normal and crashing runs

The buggy module accepts any text as an SRN without checking it (30/40 is "normalized" to 30/40). Entering ppapp as the attendance ratio makes attendance_from_ratio crash with a **ValueError** (not enough values to unpack), because it assumes the input always contains exactly one '/'.



Screenshot 2: Tests failing on the buggy module

On the original module, all 3 Hypothesis tests fail. Hypothesis reports minimal failing inputs such as None (AttributeError in all three functions) and the empty string (unpack ValueError in attendance_from_ratio, ZeroDivisionError in attendance_from_register).

```
PS C:\Users\rehan\OneDrive\Desktop\New folder\L6_L7_62_attendance_eligibility\LAB7> py -m pip install hypothesis pytest
>> py -m pytest test_attendance_eligibility_parser.py
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\rehan\OneDrive\Desktop\New folder\L6_L7_62_attendance_eligibility\LAB7\test_attendance_eligibility_parser.py:25
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\rehan\OneDrive\Desktop\New folder\L6_L7_62_attendance_eligibility\LAB7\test_attendance_eligibility_parser.py", line 23, in test_attendance_from_regi
ster_no_crash
    |     result = attendance_from_register(s)
    |   File "C:\Users\rehan\OneDrive\Desktop\New folder\L6_L7_62_attendance_eligibility\LAB7\attendance_eligibility_parser.py", line 21, in attendance_from_register
    |     marks = marks.strip().upper()
    |             ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\rehan\OneDrive\Desktop\New folder\L6_L7_62_attendance_eligibility\LAB7\test_attendance_eligibility_parser.py", line 26, in test_attendance_from_regi
ster_no_crash
    |     assert False, f"attendance_from_register crashed with: {e!r} on input {s!r}"
    | AssertionError: attendance_from_register crashed with: AttributeError("'NoneType' object has no attribute 'strip'") on input None
    | assert False
    | Failing test case: test_attendance_from_register_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\rehan\OneDrive\Desktop\New folder\L6_L7_62_attendance_eligibility\LAB7\test_attendance_eligibility_parser.py:25
    +------------------------------------
================== short test summary info ==================
FAILED test_attendance_eligibility_parser.py::test_normalize_srn_no_crash - AssertionError: normalize_srn crashed with: AttributeError(...
FAILED test_attendance_eligibility_parser.py::test_attendance_from_ratio_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 su...
FAILED test_attendance_eligibility_parser.py::test_attendance_from_register_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 su...
==================== 3 failed in 3.66s =====================
PS C:\Users\rehan\OneDrive\Desktop\New folder\L6_L7_62_attendance_eligibility\LAB7>
```

Screenshot 3: Fixed application output

The same kinds of bad input that crashed the original are now handled: the SRN PES2UG25CS820 is normalized, the ratio 30/0 (zero classes held) is reported as Invalid input instead of a ZeroDivisionError, and the register PPXAP (contains a character other than P/A) is reported as Invalid input.

```
PS C:\Users\rehan\Onpy attendance_eligibility_parser.pyattendance_eligibility\LAB7>
==== Running normalize_srn ====
Enter an SRN (e.g. pes1ug23cs001): PES2UG25CS820
Normalized SRN: PES2UG25CS820

==== Running attendance_from_ratio ====
Enter classes attended and held as ATTENDED/HELD (e.g. 30/40): 30/0
Attendance percent: Invalid input

==== Running attendance_from_register ====
Enter the register as P/A marks (e.g. PPAPP): PPXAP
Attendance percent: Invalid input
```

Screenshot 4: All tests passing

After the fixes, all 3 Hypothesis tests pass, so none of the generated inputs crash the module and every percentage returned is None or between 0 and 100.

```
PS C:\Users\rehan\OneDrive\Desktop\New folder\L6_L7_62_attendance_eligibility\LAB7> py -m pytest test_attendance_eligibility_parser.pY
==================== test session starts ====================
platform win32 -- Python 3.14.5, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\rehan\OneDrive\Desktop\New folder\L6_L7_62_attendance_eligibility\LAB7
plugins: anyio-4.13.0, hypothesis-6.170.15, asyncio-1.4.0, cov-7.1.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 3 items

test_attendance_eligibility_parser.py ...                [100%]

==================== 3 passed in 0.63s =====================
PS C:\Users\rehan\OneDrive\Desktop\New folder\L6_L7_62_attendance_eligibility\LAB7>
```

Bugs found and fixes

Function	Failing input	Error	Fix
normalize_srn	None	AttributeError	Return None if input is not a string or is empty after cleaning; remove all whitespace, not only spaces
attendance_from_ratio	None	AttributeError	Type check, return None
attendance_from_ratio	'' , '1/2/3'	ValueError (unpack)	Require exactly one '/'
attendance_from_ratio	'/' , 'abc/40'	ValueError (int)	Accept only plain ASCII digits, max 9 digits (also blocks +5, -5, 1_0, Unicode digits and huge-number DoS)
attendance_from_ratio	'30/0'	ZeroDivisionError	Reject held = 0
attendance_from_ratio	'41/40'	Wrong result (102%)	Reject attended > held
attendance_from_register	'' , ' '	ZeroDivisionError	Reject empty register
attendance_from_register	None	AttributeError	Type check, return None
attendance_from_register	'PPXP'	Wrong result	Reject characters other than P/A